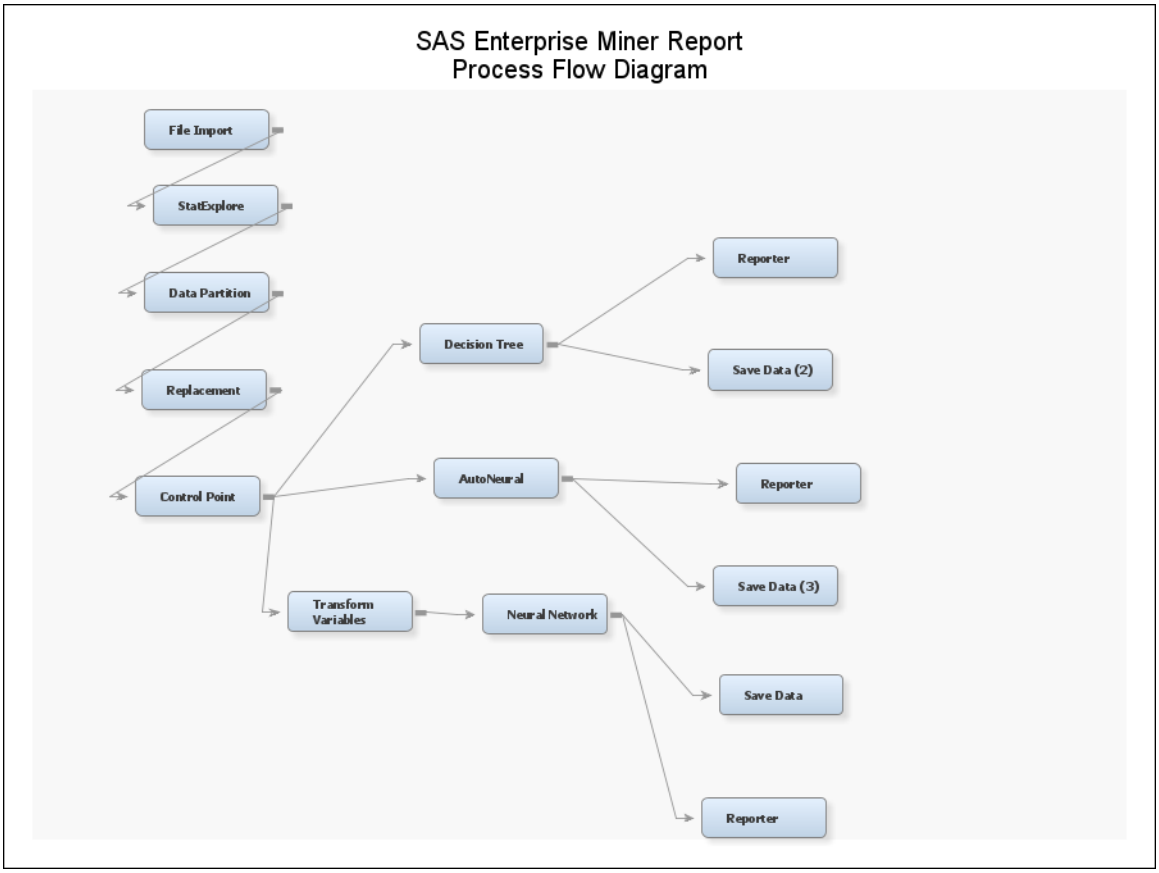
SAS Enterprise Miner Report

User = s198134
Date = 21:57:43 April 20
Project = AMS4640 (Car)
Diagram = 2023 04 18 2301

Start Node = Report3
Node label = Reporter
Nodes = ALL
Showall = N

Format = PDF
Style = LISTING



SAS Enterprise Miner Report

Node=File Import
Summary

Node id = FIMPORT
Node label = File Import
Meta path = FIMPORT
Notes =

Node=File Import
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	FileImport		GuessRows	500		NameRow	Y	
AccessTable	NoTableName		IFileName	D:\Car1.csv		Password	NoPassword	
AdvancedAdvisor	N		ImportType	Local	LOCAL	Role	TRAIN	
Delimiter	,		MaxCols	10000		SkipRows	0	
FileType	csv	XLS	MaxRows	1000000		Summarize	N	

Node=File Import
Data Attributes

Attribute	Value	Attribute	Value	Attribute	Value
Data Name	FIMPORT_DATA	Date Created	20Apr2023:21:54:48	Data Size	132096
Data Type	DATA	Date Modified	20Apr2023:21:54:48	Role	TRAIN
Data Label		Number Rows	1745	Segment	
Engine	V9	Number Columns	7	Data Library	EMWS1

Node=File Import
Variables List

Name	Label	Role	Level	Type	Length	Format	Creator
Class		TARGET	ORDINAL	N	8	BEST12.0	
buying		INPUT	ORDINAL	N	8	BEST12.0	
door		INPUT	NOMINAL	N	8	BEST12.0	
lug_boot		INPUT	ORDINAL	N	8	BEST12.0	
maint		INPUT	ORDINAL	N	8	BEST12.0	
persons		INPUT	NOMINAL	N	8	BEST12.0	
safety		INPUT	ORDINAL	N	8	BEST12.0	

Node=File Import
Created Variables List

Name	Label	Role	Type	Length
buying		INPUT	N	8
Class		TARGET	N	8
door		INPUT	N	8
lug_boot		INPUT	N	8
maint		INPUT	N	8
persons		INPUT	N	8
safety		INPUT	N	8

SAS Enterprise Miner Report

Node=StatExplore
Summary

Node id = Stat
Node label = StatExplore
Meta path = FIMPORT => Stat
Notes =

Node=StatExplore
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	StatExplore		Correlation	Y		NObs	100000	1000000
BySegment	N	Y	DropRejected	Y		Pearson	Y	
ChiSquare	Y		HideVariable	Y		Spearman	N	
ChiSquareInterval	N		IntervalDistribution	Y		UseScore	N	
ChiSquareIntervalNBins	5		LevelSummary	Y		UseTest	N	
ClassDistribution	Y		MaximumVars	1000		UseValidate	N	

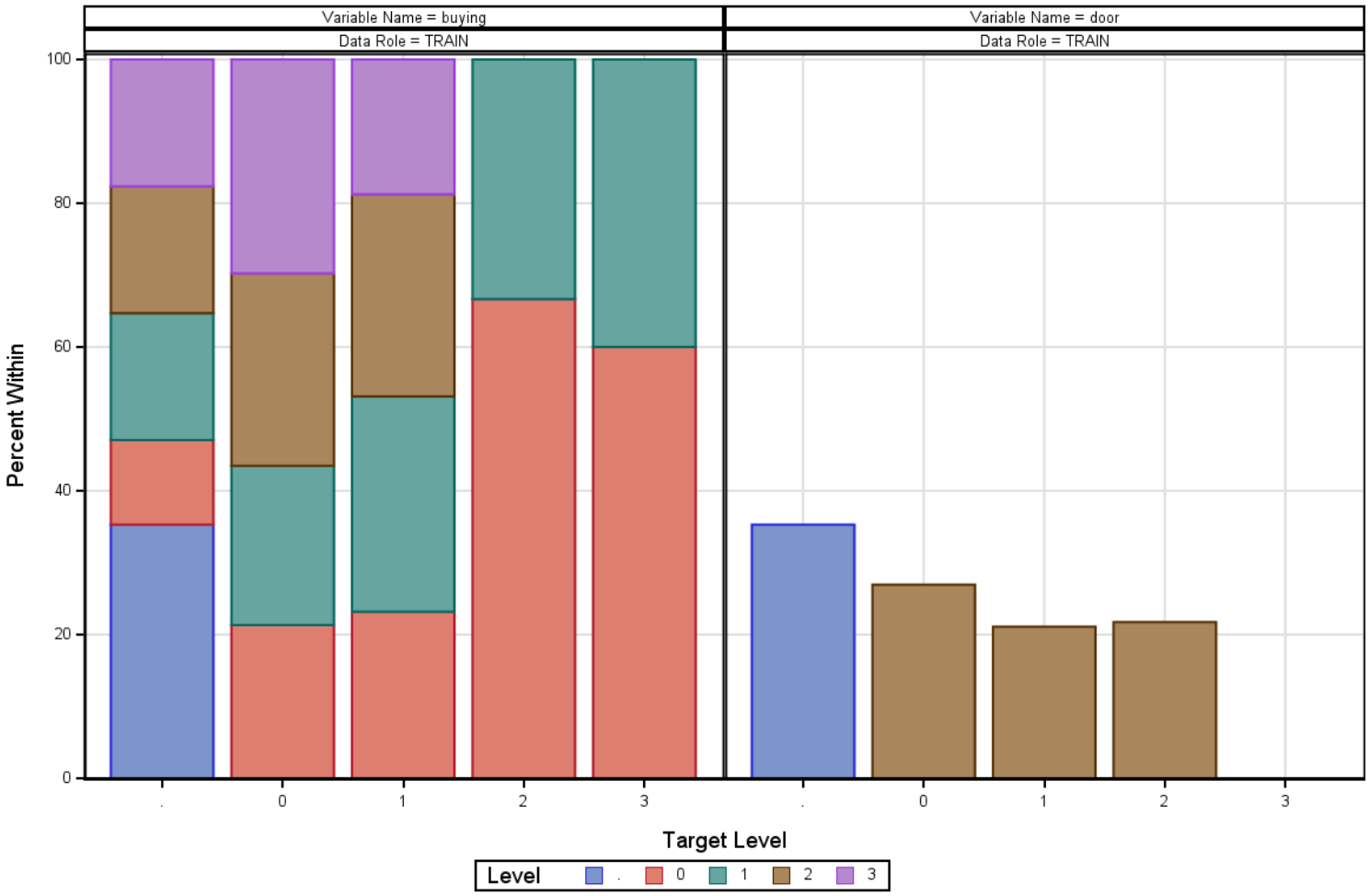
Node=StatExplore
Variable Summary

Role	Level	Frequency Count	Name
INPUT	NOMINAL	2	door persons
INPUT	ORDINAL	4	buying lug_boot maint safety

Target	Variable	Importance	Worth	Analysis Variable	Label	plot
Class	safety	1	0.15480	1	safety	.
Class	persons	2	0.09979	1	persons	.
Class	buying	3	0.02428	1	buying	.
Class	maint	4	0.01890	1	maint	.
Class	lug_boot	5	0.00855	1	lug_boot	.
Class	door	6	0.00226	1	door	.

Data Role	Segment	Segment Id	Segment Name:Value	Target	Input	Cramer's V	Prob	Chi-Square	Df	Role	Label	Ordered Inputs	Group	Plot
TRAIN			_OVERALL_	Class	safety	0.56511	<.0001	2229.0380	60	INPUT	safety	1	1	1
TRAIN			_OVERALL_	Class	persons	0.55111	<.0001	2119.9899	56	INPUT	persons	2	2	1
TRAIN			_OVERALL_	Class	door	0.50150	<.0001	1755.4869	60	INPUT	door	3	3	1
TRAIN			_OVERALL_	Class	lug_boot	0.47755	<.0001	1591.8134	48	INPUT	lug_boot	4	4	1
TRAIN			_OVERALL_	Class	buying	0.33900	<.0001	802.1508	16	INPUT	buying	5	5	1
TRAIN			_OVERALL_	Class	maint	0.32913	<.0001	756.1272	16	INPUT	maint	6	6	1

SAS Enterprise Miner Report
Node=StatExplore
Class Variables
TARGET='Class' and PLOT=1



SAS Enterprise Miner Report

Node=Data Partition
Summary

Node id = Part
Node label = Data Partition
Meta path = FIMPORT => Stat => Part
Notes =

Node=Data Partition
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Partition		Method	DEFAULT		TestPct	30	
ClassDistribution	Y		OutputType	DATA		TrainPct	70	40
IntervalDistribution	Y		RandomSeed	12345		ValidatePct	0	30

Node=Data Partition
Variable Summary

Role	Level	Frequency Count	Name
TARGET	ORDINAL	1	Class
INPUT	NOMINAL	2	door persons
INPUT	ORDINAL	4	buying lug_boot maint safety

SAS Enterprise Miner Report

Node=Replacement
Summary

Node id = Repl
Node label = Replacement
Meta path = FIMPORT => Stat => Part => Repl
Notes =

Node=Replacement
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Replace		IntervalMethod	NONE		SpacingsCutoff	9.0	9
CalcMethod	STDDEV		MADSCutoff	9.0	9	StddevCutoff	3.0	3
CountReport	Y		PercentsCutoff	0.5		UnknownLevel	NONE	MODE
HideVariable	N		ReplaceMethod	COMPUTED				

Node=Replacement
Variable Summary

Role	Level	Frequency Count	Name
INPUT	NOMINAL	2	door persons
INPUT	ORDINAL	4	buying lug_boot maint safety

SAS Enterprise Miner Report

Node=Transform Variables Summary

Node id = Trans
Node label = Transform Variables
Meta path = FIMPORT => Stat => Part => Repl => Trans
Notes =

Node=Transform Variables Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Transform		EmSampleSize	DEFAULT		MissingValue	USEINSEARCH	
DefaultClassMethod	NONE		GroupCutoff	0.1		NumberofBins	VARIABLES	
DefaultClassTargetMethod	NONE		GroupMissing	N		Offset	1	
DefaultMethod	NONE		HideVariable	Y		RejectVariable	Y	
DefaultTargetMethod	NONE		MaxOptimalBins	4		SummaryStatistics	Y	
EmRandomSeed	12345		MinOffset	Y		SummaryVariables	TRANSFORMED	
EmSampleMethod	FIRSTN		MissingAsLevel	N		UseMetaTransform	Y	

Node=Transform Variables Variable Summary

Role	Level	Frequency Count	Name
TARGET	ORDINAL	1	Class
INPUT	NOMINAL	2	door persons
INPUT	ORDINAL	4	buying lug_boot maint safety

SAS Enterprise Miner Report

Node=Neural Network
Summary

Node id = Neural
Node label = Neural Network
Meta path = FIMPORT => Stat => Part => Repl => Trans => Neural
Notes =

Node=Neural Network
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	NeuralNetwork		Hidden	3		Prelim	Y	
AbsConvValue	-1.34078E154	-7.237006E75	HiddenActivation	DEFAULT		PrelimMaxTime	1 HOUR	
AbsFTime	1		HiddenBias	Y		PrelimMaxiter	10	
AbsFValue	0		HiddenCombFunction	DEFAULT		PrelimOutest		
AbsGTime	1		HiddenUnits	N		PreliminaryRuns	5	
AbsGValue	0.00001		InitialDs			RandDist	NORMAL	
AbsXTime	1		InitialSeed	12345		RandLoc	0	
AbsXValue	1E-8		InputStandardization	STD		RandScale	0.1	
Accelerate	1.2		Learn	0.1		Residuals	Y	
AddHidden	Y		MaxLearn	50		Standardizations	N	
CodefileNoRes			MaxMomentum	1.75		SuppressOutput	N	
CodefileRes			Maxiter	50		TargetActivation	DEFAULT	
ConvDefaults	Y		Maxtime	4 HOURS		TargetBias	Y	
Decelerate	0.5		MinLearn	0.00001		TargetCombFunction	DEFAULT	
DirectConnection	Y	N	ModelSelectionCriterion	PROFIT/LOSS		TargetError	DEFAULT	
FConvTime	1		Momentum	0		Tilt	0	
FConvValue	0		NetworkArchitecture	MLP		TrainCode		
GConvTime	1		Outest			TrainingTechnique	BPROP	DEFAULT
GConvValue	1E-6		Outfit			UseEstimates	N	

Node=Neural Network
Variable Summary

Role	Level	Frequency Count	Name
TARGET	ORDINAL	1	Class
INPUT	NOMINAL	2	door persons
INPUT	ORDINAL	4	buying lug_boot maint safety

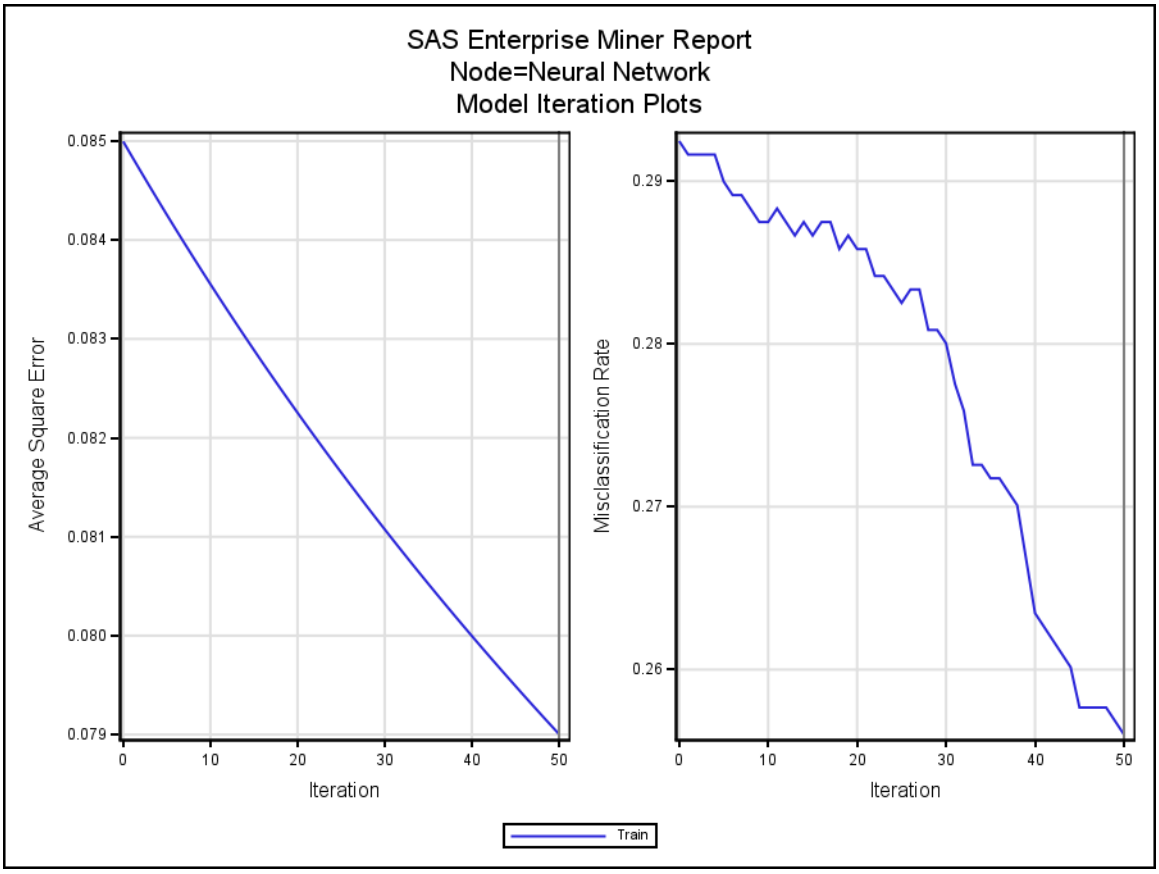
Node=Neural Network
Model Fit Statistics

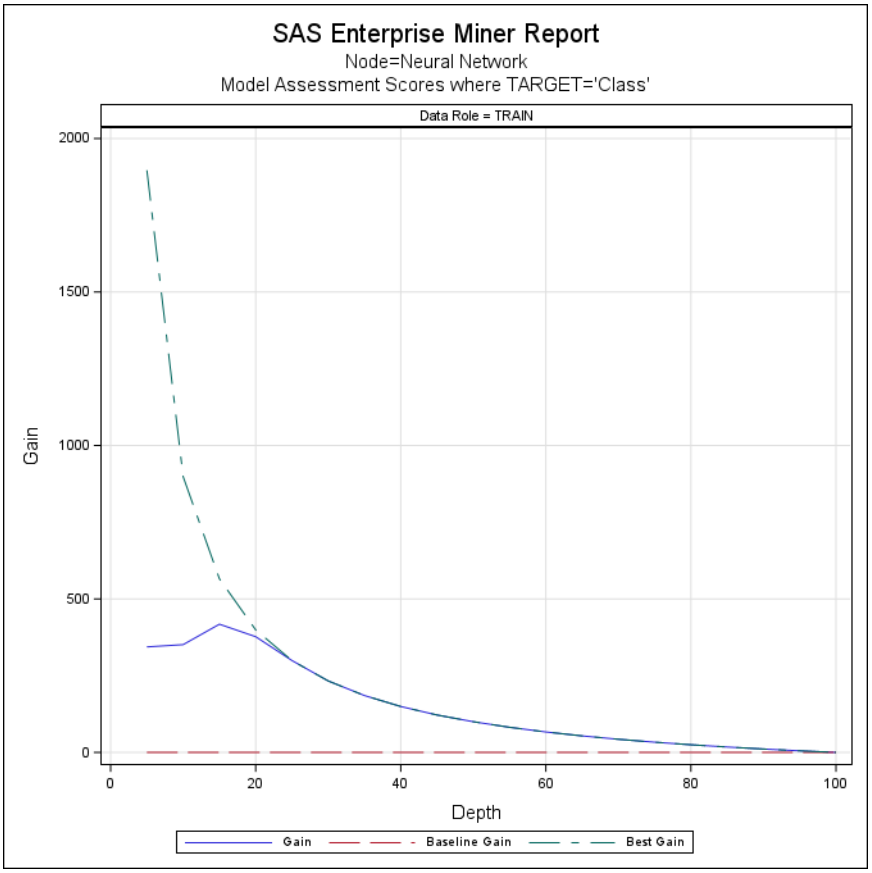
Target=Class Target Label=' '

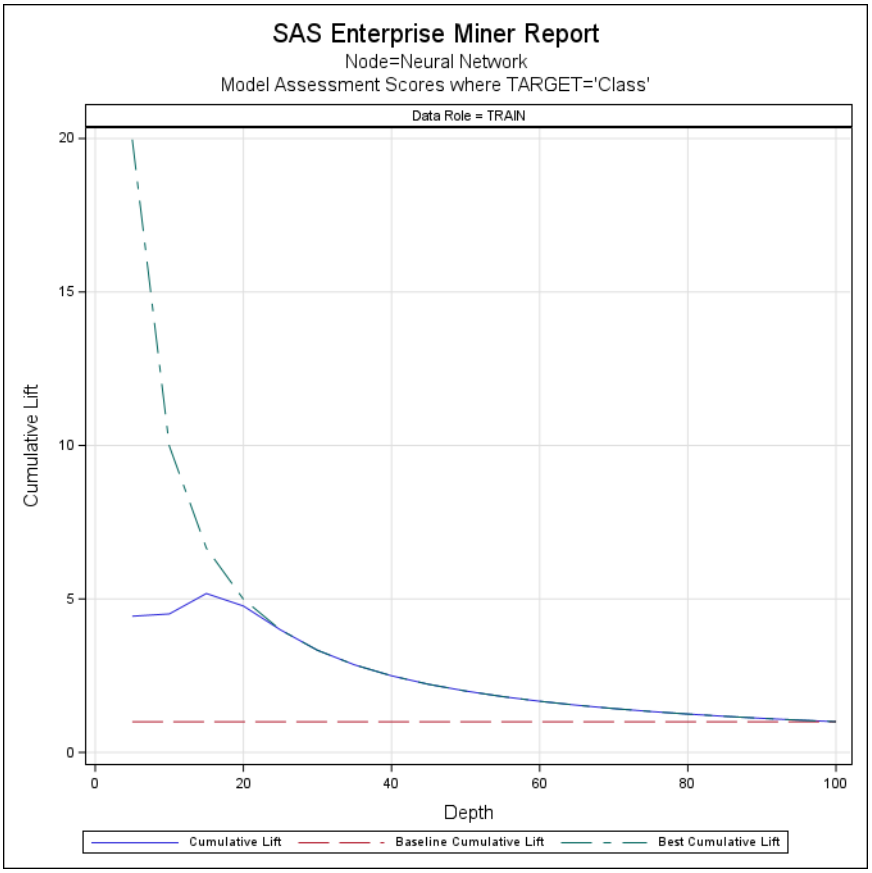
Label of Statistic	Train	Validation	Test
Total Degrees of Freedom	3621.00	.	.
Degrees of Freedom for Error	3416.00	.	.
Model Degrees of Freedom	205.00	.	.
Number of Estimated Weights	205.00	.	.
Akaike's Information Criterion	1874.66	.	.

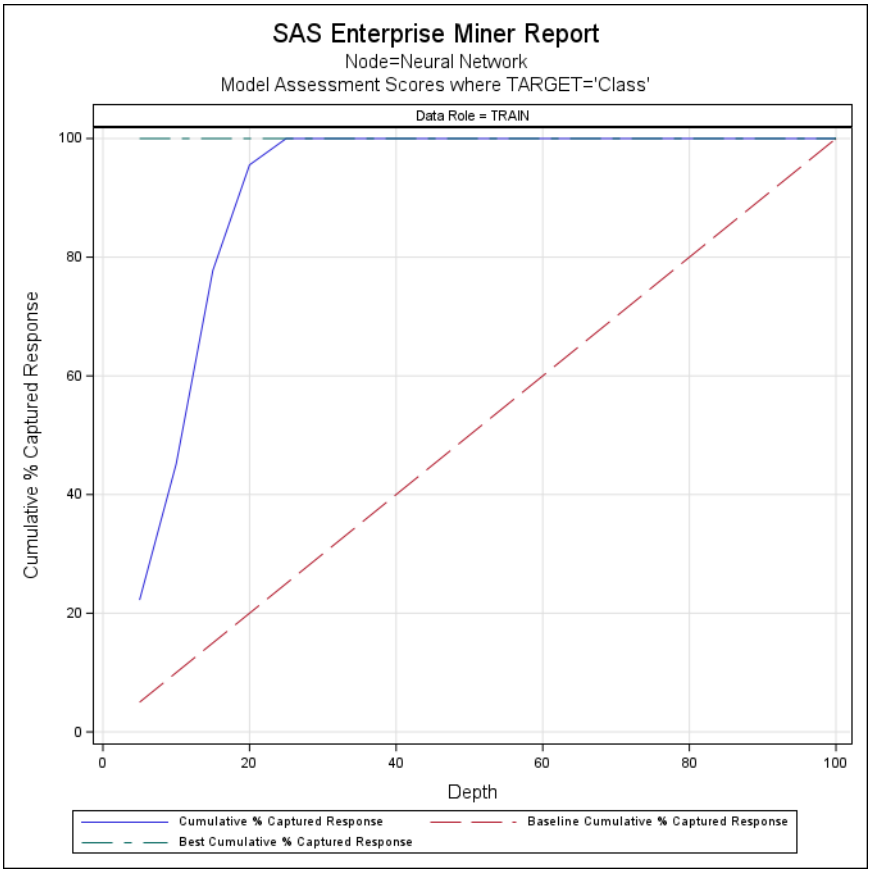
Target=Class Target Label=' '

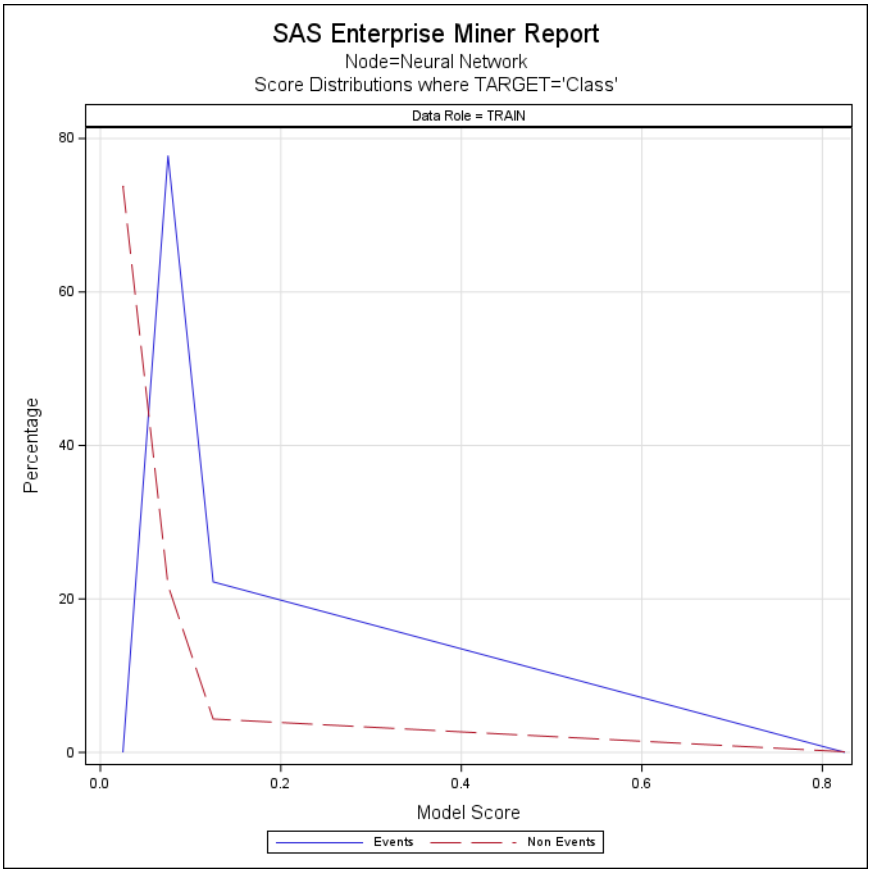
Label of Statistic	Train	Validation	Test
Schwarz's Bayesian Criterion	3144.53	.	.
Average Squared Error	0.08	.	0.08
Maximum Absolute Error	0.96	.	0.96
Divisor for ASE	4828.00	.	2084.00
Sum of Frequencies	1207.00	.	521.00
Root Average Squared Error	0.28	.	0.28
Sum of Squared Errors	381.43	.	165.34
Sum of Case Weights Times Freq	4828.00	.	2084.00
Final Prediction Error	0.09	.	.
Mean Squared Error	0.08	.	0.08
Root Final Prediction Error	0.30	.	.
Root Mean Squared Error	0.29	.	0.28
Average Error Function	0.30	.	0.31
Error Function	1464.66	.	637.51
Misclassification Rate	0.26	.	0.26
Number of Wrong Classifications	309.00	.	135.00

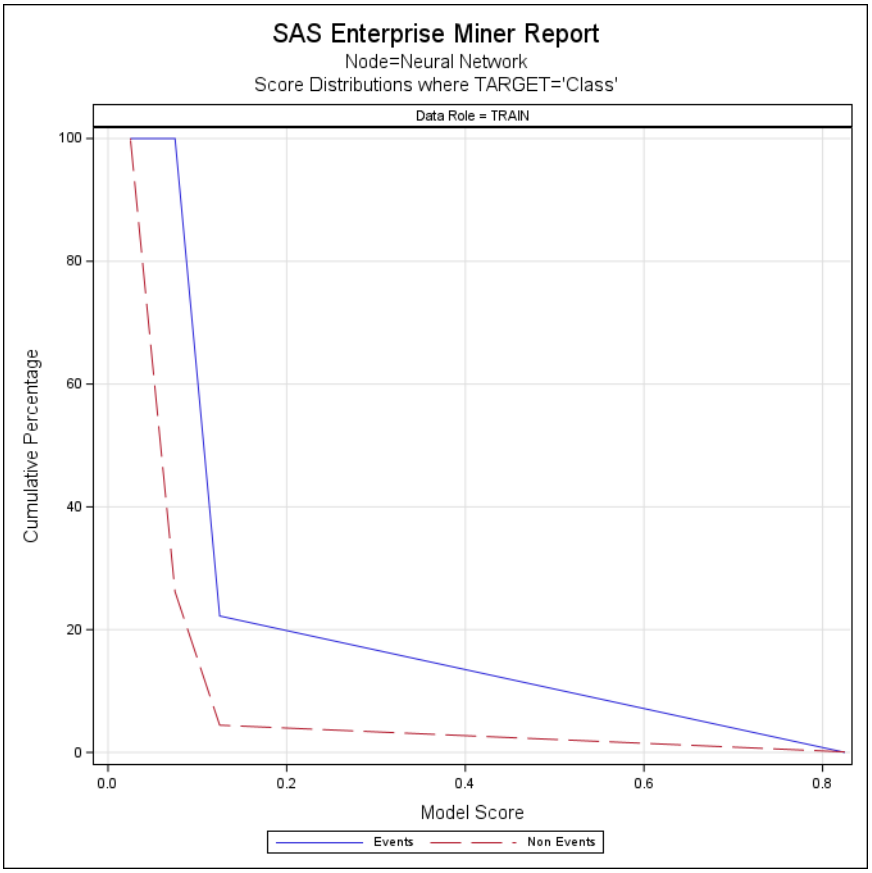



SAS Enterprise Miner Report
Node=Neural Network
Model Assessment Scores where TARGET='Class'
Data Role = TRAIN
Gain
2000
1500
1000
500
0
0
20
40
60
80
100
Depth
Gain
Baseline Gain
Best Gain


SAS Enterprise Miner Report
Node=Neural Network
Model Assessment Scores where TARGET='Class'
Data Role = TRAIN
Cumulative Lift
20
15
10
5
0
0
20
40
60
80
100
Depth
Cumulative Lift
Baseline Cumulative Lift
Best Cumulative Lift


SAS Enterprise Miner Report
Node=Neural Network
Model Assessment Scores where TARGET='Class'
Data Role = TRAIN
Cumulative % Captured Response
Depth
0
20
40
60
80
100
Cumulative % Captured Response
Baseline Cumulative % Captured Response
Best Cumulative % Captured Response


SAS Enterprise Miner Report
Node=Neural Network
Score Distributions where TARGET='Class'
Data Role = TRAIN
Percentage
Model Score
0
20
40
60
80
0.0
0.2
0.4
0.6
0.8
Events
Non Events



Node=Neural Network
Score Distributions

Target Variable=Class Data Role=TRAIN

Posterior Probability Range	Number of Events	Percentage of Events	Percentage of Nonevents	Cumulative Percentage of Events	Cumulative Percentage of Nonevents
0.80-0.85	0	0.0000	0.0853	0.000	0.085
0.10-0.15	10	22.2222	4.3478	22.222	4.433
0.05-0.10	35	77.7778	21.7391	100.000	26.172
0.00-0.05	0	0.0000	73.8278	100.000	100.000

SAS Enterprise Miner Report

Node=AutoNeural
Summary

Node id = AutoNeural
Node label = AutoNeural
Meta path = FIMPORT => Stat => Part => Repl => AutoNeural
Notes =

Node=AutoNeural
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	AutoNeural		HistoryDs			Tanh	Y	
AdjustIterations	Y		Identity	N		TargetError	DEFAULT	
Architecture	SINGLE LAYER		Logistic	N		Termination	OVERFITTING	
Bestds			MaxIter	8		Tolerance	MEDIUM	
Direct	Y		Normal	Y		TotalHidden	30	
Exponential	N		Reciprocal	N		TotalTime	1 HOUR	1 Hour
FinalIter	5		Residuals	Y		TrainAction	SEARCH	
FinalTrain	Y		Sine	Y		estds		
Freeze	N		Softmax	N		outfit		
Hidden	2		Square	N		weights		
HiddenUnits	N		Standardizations	N				

Node=AutoNeural
Variable Summary

Role	Level	Frequency Count	Name
TARGET	ORDINAL	1	Class
INPUT	NOMINAL	2	door persons
INPUT	ORDINAL	4	buying lug_boot maint safety

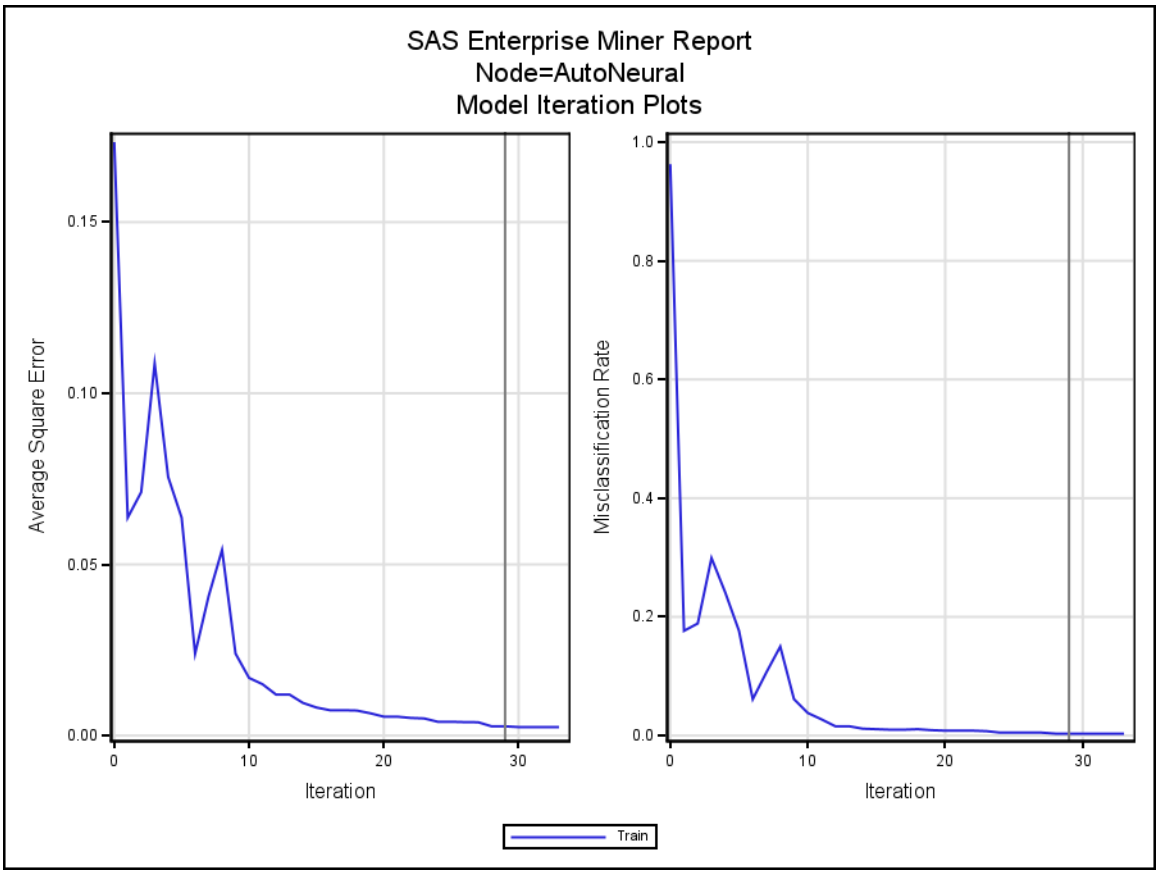
Node=AutoNeural
Model Fit Statistics

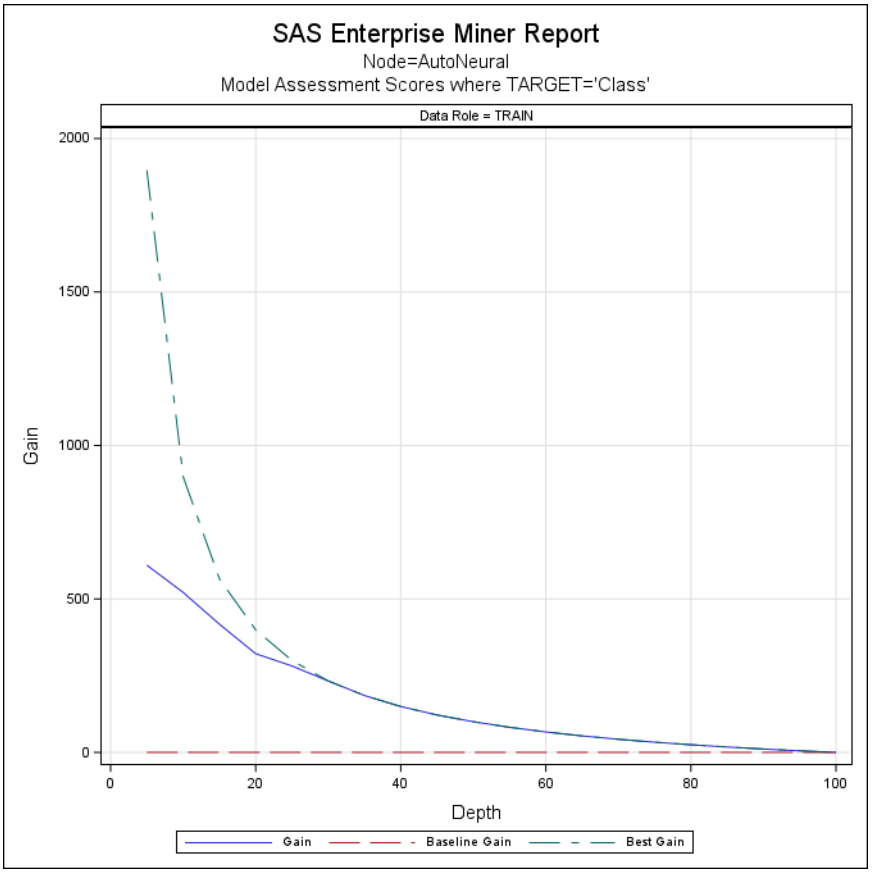
Target=Class Target Label=' '

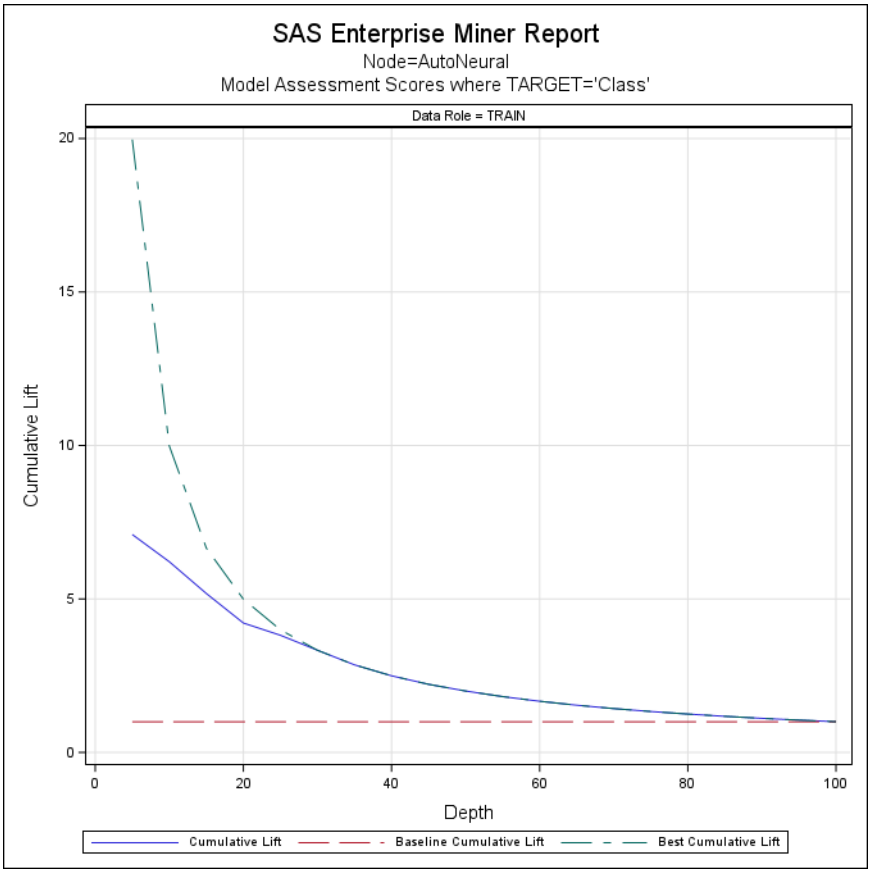
Label of Statistic	Train	Validation	Test
Total Degrees of Freedom	3621.00	.	.
Degrees of Freedom for Error	3006.00	.	.
Model Degrees of Freedom	615.00	.	.
Number of Estimated Weights	615.00	.	.
Akaike's Information Criterion	13665.93	.	.
Schwarz's Bayesian Criterion	17475.55	.	.
Average Squared Error	0.29	.	0.29
Maximum Absolute Error	1.00	.	1.00
Divisor for ASE	4828.00	.	2084.00
Sum of Frequencies	1207.00	.	521.00
Root Average Squared Error	0.53	.	0.54
Sum of Squared Errors	1381.83	.	596.82
Sum of Case Weights Times Freq	4828.00	.	2084.00

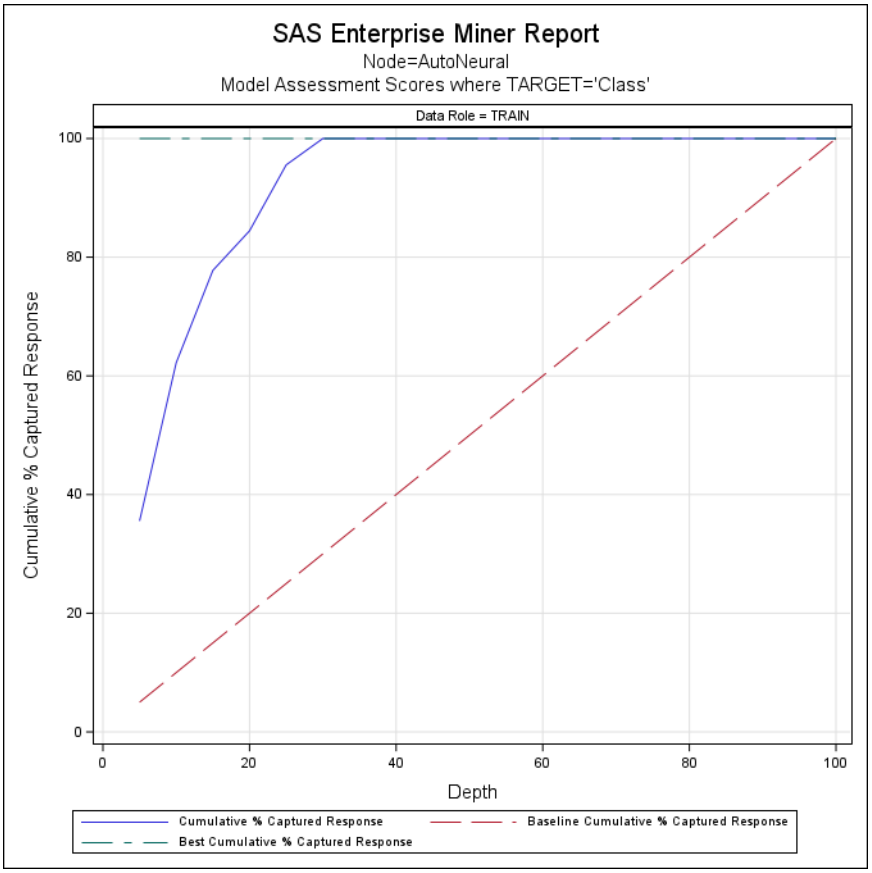
Target=Class Target Label=' '

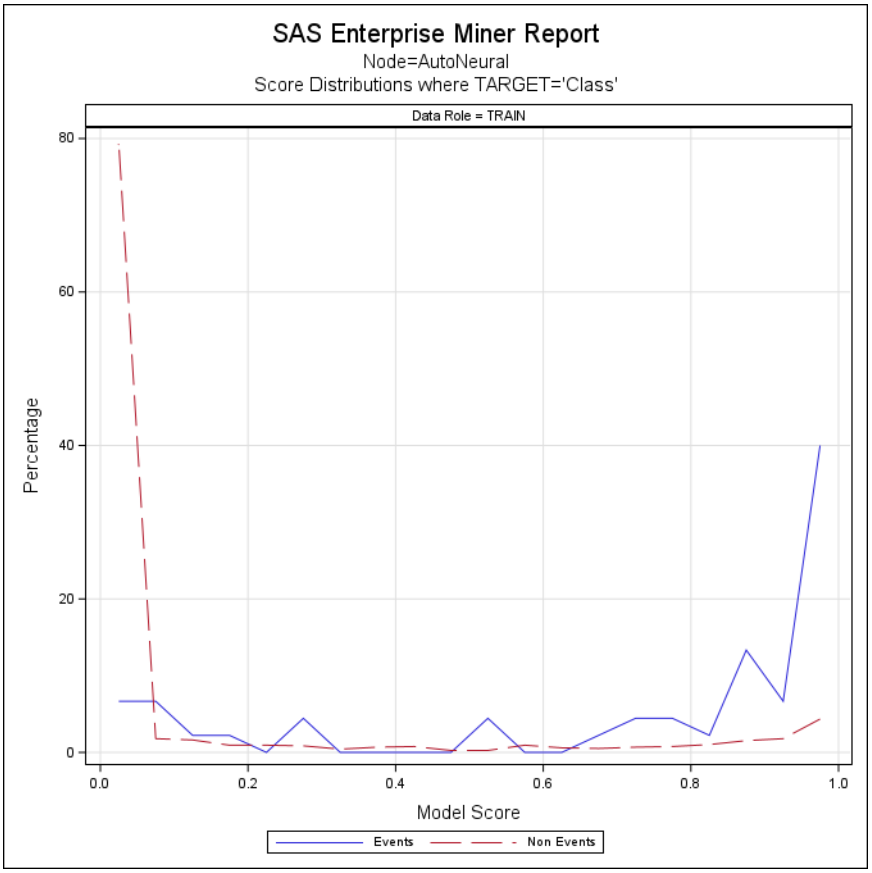
Label of Statistic	Train	Validation	Test
Final Prediction Error	0.40	.	.
Mean Squared Error	0.34	.	0.29
Root Final Prediction Error	0.64	.	.
Root Mean Squared Error	0.59	.	0.54
Average Error Function	2.58	.	2.67
Error Function	12435.93	.	5572.18
Misclassification Rate	0.67	.	0.68
Number of Wrong Classifications	804.00	.	354.00

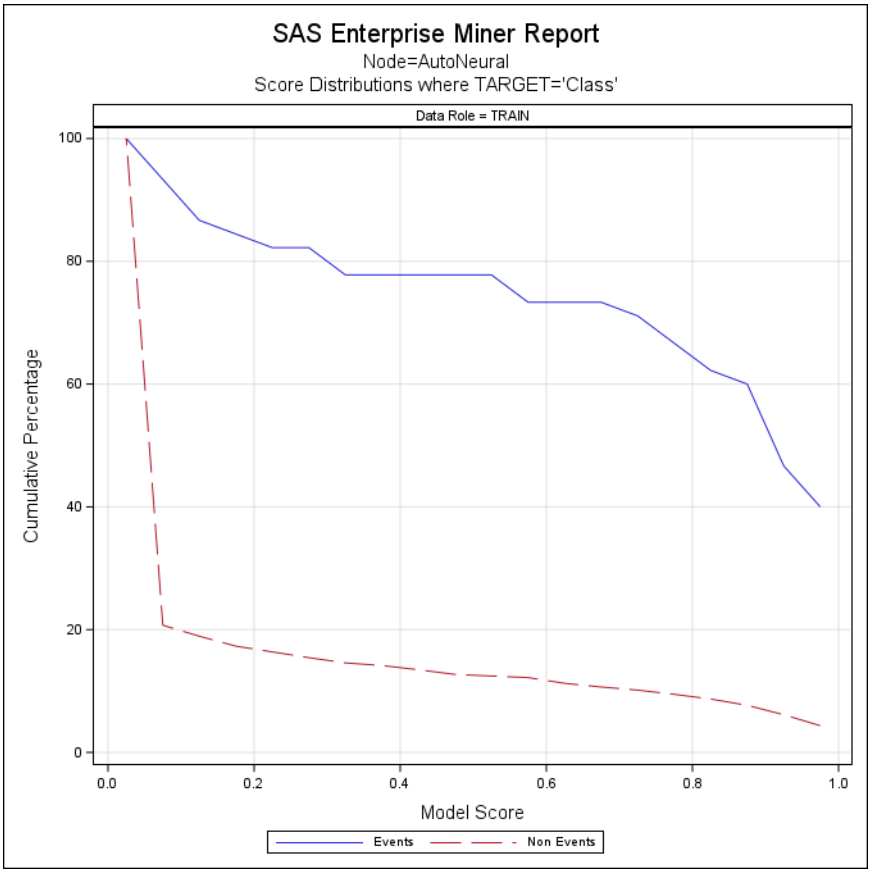



SAS Enterprise Miner Report
Node=AutoNeural
Model Assessment Scores where TARGET='Class'
Data Role = TRAIN
Gain
2000
1500
1000
500
0
0
20
40
60
80
100
Depth
Gain
Baseline Gain
Best Gain


SAS Enterprise Miner Report
Node=AutoNeural
Model Assessment Scores where TARGET='Class'
Data Role = TRAIN
Cumulative Lift
20
15
10
5
0
0
20
40
60
80
100
Depth
Cumulative Lift
Baseline Cumulative Lift
Best Cumulative Lift


SAS Enterprise Miner Report
Node=AutoNeural
Model Assessment Scores where TARGET='Class'
Data Role = TRAIN
Cumulative % Captured Response
Depth
0
20
40
60
80
100
Cumulative % Captured Response
Baseline Cumulative % Captured Response
Best Cumulative % Captured Response


SAS Enterprise Miner Report
Node=AutoNeural
Score Distributions where TARGET='Class'
Data Role = TRAIN
Percentage
Model Score
0
20
40
60
80
0.0
0.2
0.4
0.6
0.8
1.0
Events
Non Events



Node=AutoNeural
Score Distributions

Target Variable=Class Data Role=TRAIN

Posterior Probability Range	Number of Events	Percentage of Events	Percentage of Nonevents	Cumulative Percentage of Events	Cumulative Percentage of Nonevents
0.95-1.00	18	40.0000	4.3478	40.000	4.348
0.90-0.95	3	6.6667	1.7903	46.667	6.138
0.85-0.90	6	13.3333	1.5345	60.000	7.673
0.80-0.85	1	2.2222	1.0230	62.222	8.696
0.75-0.80	2	4.4444	0.7673	66.667	9.463
0.70-0.75	2	4.4444	0.6820	71.111	10.145
0.65-0.70	1	2.2222	0.5115	73.333	10.656
0.60-0.65	0	0.0000	0.5968	73.333	11.253
0.55-0.60	0	0.0000	0.9378	73.333	12.191
0.50-0.55	2	4.4444	0.2558	77.778	12.447
0.45-0.50	0	0.0000	0.2558	77.778	12.702
0.40-0.45	0	0.0000	0.7673	77.778	13.470
0.35-0.40	0	0.0000	0.6820	77.778	14.152
0.30-0.35	0	0.0000	0.4263	77.778	14.578
0.25-0.30	2	4.4444	0.8525	82.222	15.431
0.20-0.25	0	0.0000	0.9378	82.222	16.368
0.15-0.20	1	2.2222	0.9378	84.444	17.306
0.10-0.15	1	2.2222	1.6198	86.667	18.926
0.05-0.10	3	6.6667	1.7903	93.333	20.716
0.00-0.05	3	6.6667	79.2839	100.000	100.000

End of Report